

$$T = 2 \text{ mg}$$

- 12 D** Since net gravitational force is zero,

$$\frac{GM_1m}{r_1^2} = \frac{GM_2m}{r_2^2}$$

$$\frac{9M}{(15-x)^2} = \frac{4M}{x^2}$$

$$\frac{3}{15-x} = \frac{2}{x}$$

$$3x = 30 - 2x$$

$$x = 6.0 \text{ m}$$

